

Chencheng Liang

PhD in Computer Science

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🗣️ ChenchengLiang

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Work Experience

2025-08 – 2026-06

Independent AI & Automation Consultant

- Built an LLM-based web scraping system to collect and consolidate product information from online retailers and supplier sources, using web scraping, image-based information extraction, structured outputs, and multilingual data standardization
- Developed an AI-assisted independent e-commerce store on Shopify, using LLM workflows for product catalog creation, attribute extraction, product categorization, SEO metadata generation, multilingual localization, and marketing content generation
- Delivered AI automation projects and hands-on training for small business owners, covering prompt engineering, LLM tool selection, workflow integration, and business-process automation

Education

2019 – 2025

Uppsala University, Sweden

PhD in Computer Science

Thesis: “Learning to Guide Automated Reasoning: A GNN-Based Framework”

Advisors: Philipp Rümmer, Yi Wang, Marc Brockschmidt

Opponent: Stephan Schulz

2016 – 2018

University of Göttingen, Germany

Master of Science in Computer Science

Thesis: “Topology Control Using Fuzzy Game Theory in Mobile Underwater Sensor Networks”

Grade: 1.0 (Sehr gut, highest distinction)

2010 – 2014

Chengdu University, China

Bachelor of Science in Software Engineering

Graduated 1st in class

Teaching Experience

2019 – 2025

Uppsala University, Sweden

Course Instructor & Teaching Assistant

- **Natural Computation Methods for Machine Learning** (Master’s, 10 credits, 2019-2025)
- **Imperative and Object-Oriented Programming Methodology** (Bachelor’s, 10 credits, 2023-2024)
- **Project CS** (Master’s project course, 30 credits, 2022-2024)
- **Introduction to Studies in Embedded Systems** (Master’s, 5 credits, 2019-2021)

Thesis & Project Supervision

2024	Federated Learning meets Large Language Models <i>Master's Project Supervision</i> Students: Mahtab Mirhaj, Oguzhan Ersoy, Yixin Huang, Shiqi Shu Ericsson Research & Uppsala University
2023	Distributed Mobility Prediction in Telecommunication Networks <i>Master's Project Supervision</i> Students: Shreyansh Singh, Lifang Zheng, Savvas Giortsis, Tobias Lass, Yilei Zheng, Salum Nassor Ericsson Research & Uppsala University
2022	Shortest Path Reliable Estimation: Applying Graph Neural Networks - SpreadNet <i>Master's Project Supervision</i> Students: ChanVuth Chea, Boli Gao, Jennifer Gross, Haouyuan Li, Paarth Sanhotra, George-Alexandru Stoian, Sofia Afnaan Syed, Haodong Zhao Ericsson Research & Uppsala University
2021	Exploring Properties and Limitations of Graph Neural Networks (GNNs) in Software Verification <i>Master's Thesis Supervision</i> Student: Kexin Xu Uppsala University

Notable Projects & Achievements

2025	CozySteps - LLM-powered Activity Recommender <i>Impact Hack 2025, AI Sweden</i> - Built mobile app prototype for families with autistic children - Implemented LLM-based activity recommendation engine - Won recognition for addressing real societal challenges
2023	Insilico - AI-Driven Art Installation <i>Uppsala Art Museum</i> - Integrated AI-based sentiment analysis for Mat Collishaw's art installation - Developed real-time Twitter sentiment analysis system - Connected digital hostility to physical art expression

Research Interests & Skills

- **Primary Research Areas:** Neural network training, Graph Neural Networks (GNNs), Symbolic reasoning, AI workflow automation
- **Technical Skills:** Python, Scala, Java, C/C++, PyTorch, TensorFlow, Git, Linux, Docker
- **AI/ML:** Deep Learning (CNN, RNN, GNN, Transformers), Reinforcement Learning, NLP, Classical ML
- **Languages:** Chinese (Native), English (Fluent), Japanese (Basic)

Professional Service

- **Subreviewer for Conferences:**
 - FMCAD 2025, FM 2024, LPAR 2024, VMCAI 2024, FMCAD 2023, LPAR 2023
 - FMCAD 2022, IJCAR 2022, FroCoS 2021, CADE 2021, TACAS 2020
 - iFM 2019, FroCoS 2019, CADE 2019

- **Talks & Presentations:**

- “Guiding Constraint Horn Clauses Solving using Graph Neural Networks” - ISTA, Austria, 2024
- “Boosting Constrained Horn Solving by Unsat Core Learning” - HCVS Workshop, Luxembourg, 2024
- “Exploring Representation of Horn Clauses using GNNs” - AITP Conference, France, 2022
- “Guiding Interpolation for Model Checking by Deep Learning Techniques” - AVM, Czech Republic, 2019

Publications

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| 2025 | 1. Abdulla, P. A., Atig, M. F., Cailler, J., Liang, C. & Rümmer, P. <i>When GNNs Met a Word Equations Solver: Learning to Rank Equations</i> in <i>15th International Symposium on Frontiers of Combining Systems (FroCoS)</i> (Springer, 2025). |
| 2024 | 2. Abdulla, P. A., Atig, M. F., Cailler, J., Liang, C. & Rümmer, P. <i>Guiding Word Equation Solving Using Graph Neural Networks</i> in <i>22nd International Symposium on Automated Technology for Verification and Analysis (ATVA)</i> (Springer, 2024). |
| | 3. Abdulla, P. A., Liang, C. & Rümmer, P. <i>Boosting Constrained Horn Solving by Unsat Core Learning</i> in <i>25th International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI)</i> (Springer, 2024). |
| 2022 | 4. Liang, C. , Rümmer, P. & Brockschmidt, M. <i>Exploring Representation of Horn clauses using GNNs</i> in <i>8th Workshop on Practical Aspects of Automated Reasoning (PAAR)</i> (2022). |
| 2021 | 5. Yuan, Y., Liang, C. , Chen, X., Baker, T. & Fu, X. Adaptive Fuzzy Game-Based Energy-Efficient Localization in 3D Underwater Sensor Networks. <i>ACM Transactions on Internet Technology</i> 22 , 1–20 (2021). |
| 2019 | 6. Yuan, Y., Liang, C. , Kaneko, M., Chen, X. & Hogrefe, D. Topology Control for Energy-Efficient Localization in Mobile Underwater Sensor Networks Using Stackelberg Game. <i>IEEE Transactions on Vehicular Technology</i> 68 , 1487–1500 (2019). |
| 2017 | 7. Yang, Y., Liang, C. & Ji, S. Comments on “Fuzzy multicriteria decision making method based on the improved accuracy function for interval-valued intuitionistic fuzzy sets” by Ridvan Sahin. <i>Soft Computing</i> 21 , 3033–3035 (2017). |
| 2015 | 8. Yang, Y., Liang, C. , Ji, S. & Liu, T. Adjustable soft discernibility matrix based on picture fuzzy soft sets and its applications in decision making. <i>Journal of Intelligent & Fuzzy Systems: Applications in Engineering and Technology</i> 29 , 1711–1722 (2015). |